A MORPHISM OF CONE TRIANGLES WHOSE FIRST TWO COMPONENTS ARE HOMOTOPY EQUIVALENCES HAS A HOMOTOPY EQUIVALENCE ON THE CONES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the inversion of a and b with the cone completion c' through lemma-cone, the reduction to the endomorphism case through lemma-triseq, and the difference trick with (c - 1)^2 = 0 giving the inverse 2 - c are all retained verbatim.

Editorial note. The label lemma-analogue-third-isomorphism names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Completing a morphism on cones. Same chapter, Lemma lemma-cone. Builds the inverse component c' on the cones from the inverse homotopy equivalences a, b, through the left commutative box.
- The two-out-of-three reduction for triangle morphisms. Same chapter, Lemma lemma-triseq. Reduces the assertion to the endomorphism case (1, 1, c) and yields the identity (c - 1)^2 = 0 from the admissible rows of the difference diagram.

Source reference. Original reference 08C5: sources/stacks-dga/pinned-original-source.tex, source lines 3779–3779. Prerequisite pointer invoked at proof line(s) 4285; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09QP: sources/stacks-dga/pinned-original-source.tex, source lines 4050–4050. Prerequisite pointer invoked at proof line(s) 4287; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09QJ: sources/stacks-dga/pinned-original-source.tex, source lines 3762–3762. Prerequisite pointer invoked at proof line(s) 4262; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 09QJ let $f_1 : x_1 \to y_1$ and $f_2 : x_2 \to y_2$ be morphisms in Comp($\mathcal{A}$). Let*

$$(a, b, c) : (x_1, y_1, c(f_1), f_1, i_1, p_1) \to (x_2, y_2, c(f_2), f_2, i_1, p_1)$$

be any morphism of triangles in $K(\mathcal{A})$. If a and b are homotopy equivalences, then so is c.

Proof. Since a and b are homotopy equivalences, they are invertible in $K(\mathcal{A})$ so let a^{-1} and b^{-1} denote their inverses in $K(\mathcal{A})$, giving us a commutative diagram

$$\begin{array}{ccccc} x_2 & \xrightarrow{f_2} & y_2 & \xrightarrow{i_2} & c(f_2) \\ \downarrow{\scriptstyle a^{-1}} & & \downarrow{\scriptstyle b^{-1}} & & \downarrow{\scriptstyle c'} \\ x_1 & \xrightarrow{f_1} & y_1 & \xrightarrow{i_1} & c(f_1) \end{array}$$

where the map c' is defined via Lemma 08C5 applied to the left commutative box of the above diagram. Since the diagram commutes in $K(\mathcal{A})$, it suffices by Lemma 09QP to prove the following: given a morphism of triangle $(1, 1, c) : (x, y, c(f), f, i, p) \to (x, y, c(f), f, i, p)$ in $K(\mathcal{A})$, the map c is an isomorphism in $K(\mathcal{A})$. We have the commutative diagrams in $K(\mathcal{A})$:

$$\begin{array}{ccccc} y & \longrightarrow & c(f) & \longrightarrow & x[1] \\ \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle c} & & \downarrow{\scriptstyle 1} \\ y & \longrightarrow & c(f) & \longrightarrow & x[1] \end{array} \quad \Rightarrow \quad \begin{array}{ccccc} y & \longrightarrow & c(f) & \longrightarrow & x[1] \\ \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle c-1} & & \downarrow{\scriptstyle 0} \\ y & \longrightarrow & c(f) & \longrightarrow & x[1] \end{array}$$

Since the rows are admissible short exact sequences, we obtain the identity $(c-1)^2 = 0$ by Lemma 09QP, from which we conclude that $2 - c$ is inverse to c in $K(\mathcal{A})$ so that c is an isomorphism. □